JETSMUNT

DEFENSE PROPULSION SYSTEMS

European Turbojet Propulsion
for Advanced Unmanned Systems

SINCE 1998
Research and Development Started

6,000+ ENGINES DELIVERED
Worldwide Operational Experience

DESIGNED & MANUFACTURED
In Spain

JETSMUNT
DEFENSE PROPULSION SYSTEMS

ABOUT JETSMUNT

More than three decades of turbojet propulsion expertise.

COMPLETE PRODUCT PORTFOLIO
98 N – 255 N

Jetsmunt is a Spanish aerospace engineering company specialized in the design, development and manufacturing of compact turbojet propulsion systems.

The company's origins date back to 1994, when research activities in turbine technology began. Following years of development, the first successful flight was achieved in 1996 and the company was formally established in 1998.

Today, Jetsmunt develops complete propulsion solutions combining advanced turbine technology, proprietary electronic control systems and integration capabilities for modern unmanned platforms.

With more than 6,000 engines delivered worldwide, Jetsmunt has accumulated decades of operational experience across a wide range of aerospace applications.

1994	1996	1998	2025+
Research Activities Started	First Successful Flight	Company Founded	6,000+ Engines Delivered Worldwide

PRODUCT RANGE
98 N – 255 N

GLOBAL REACH
50+ Countries

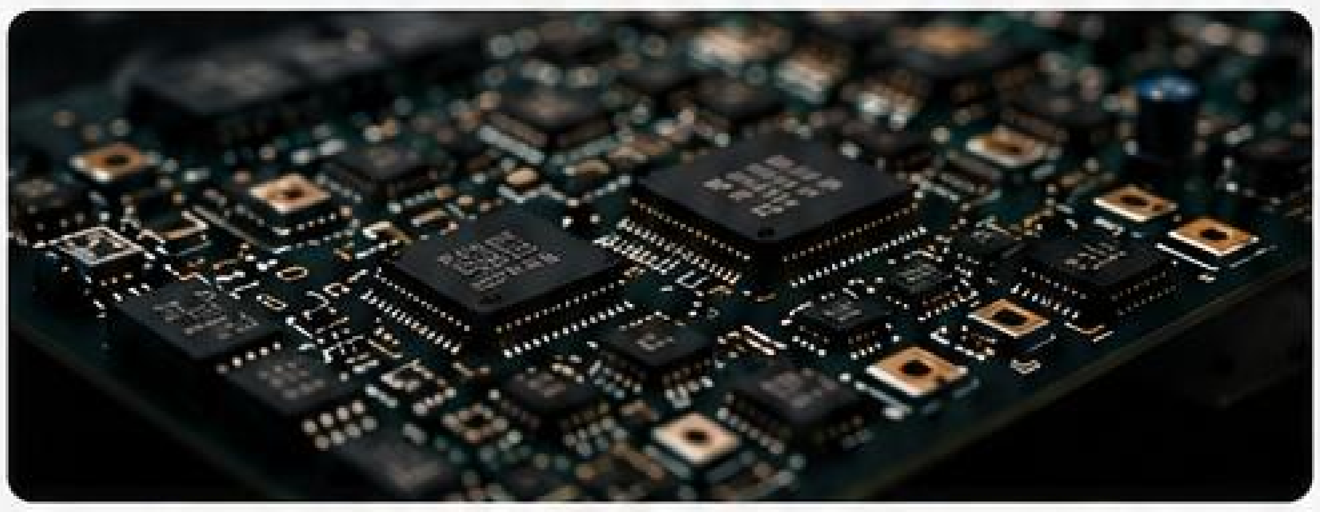
ENGINEERING EXPERIENCE
30+ Years

CORE COMPETENCIES

Complete in-house capabilities for UAV propulsion systems.

TURBINE DESIGN

In-house aerodynamic design, thermodynamic analysis and mechanical development optimized for UAV applications.



ECU DEVELOPMENT

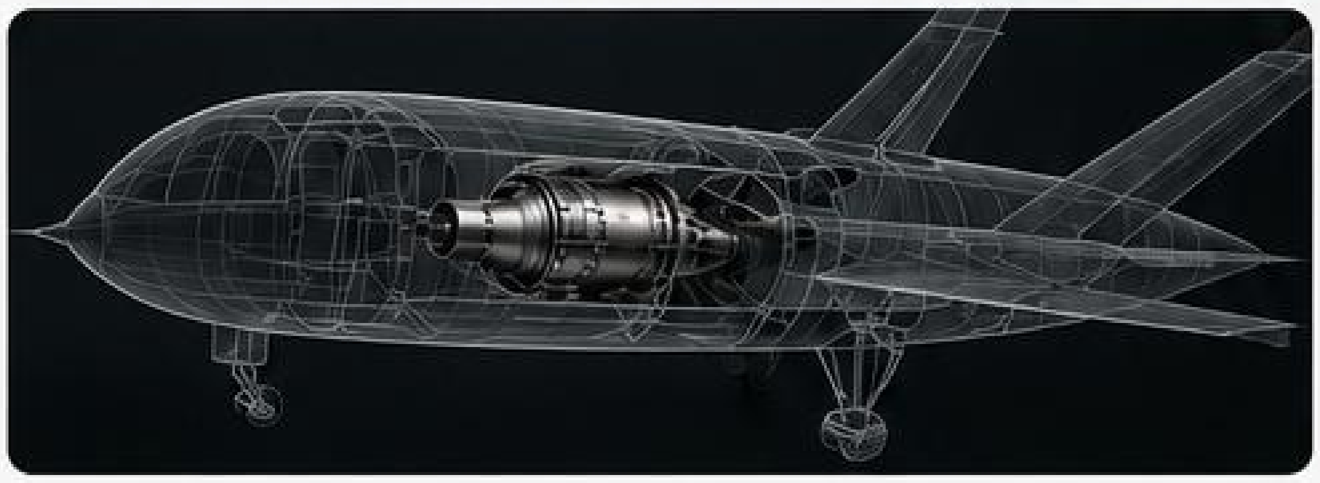
Proprietary electronic control units developed in-house for precise engine control, monitoring and protection.

EMBEDDED SOFTWARE

Real-time control algorithms, diagnostics and performance optimization tailored to each mission requirement.

TELEMETRY SYSTEMS

Engine monitoring, data acquisition and mission telemetry integration for real-time situational awareness.



UAV INTEGRATION

Propulsion integration support for OEMs, research organizations and aerospace programs. Flexible solutions for rapid integration and testing.

DESIGN

Advanced engineering and development capabilities.

TEST

In-house testing and validation for reliability and safety.

MANUFACTURE

Precision manufacturing and assembly in Spain under strict quality control.

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ABOUT JETSMUNT

EUROPEAN ENGINEERING. SPANISH MANUFACTURING.

Founded in 1998, Jetsmunt is a Spanish engineering company specialized in the design, development and manufacture of compact turbojet engines and propulsion systems for UAV and defense applications.

With more than 25 years of experience and thousands of engines delivered worldwide, we combine advanced engineering with agile processes to provide reliable, efficient and cost-effective solutions.

FOUNDED IN 1998
More than 25 years of continuous innovation in turbine technology.

25+ YEARS OF EXPERIENCE
Proven track record in the development and production of jet propulsion systems.

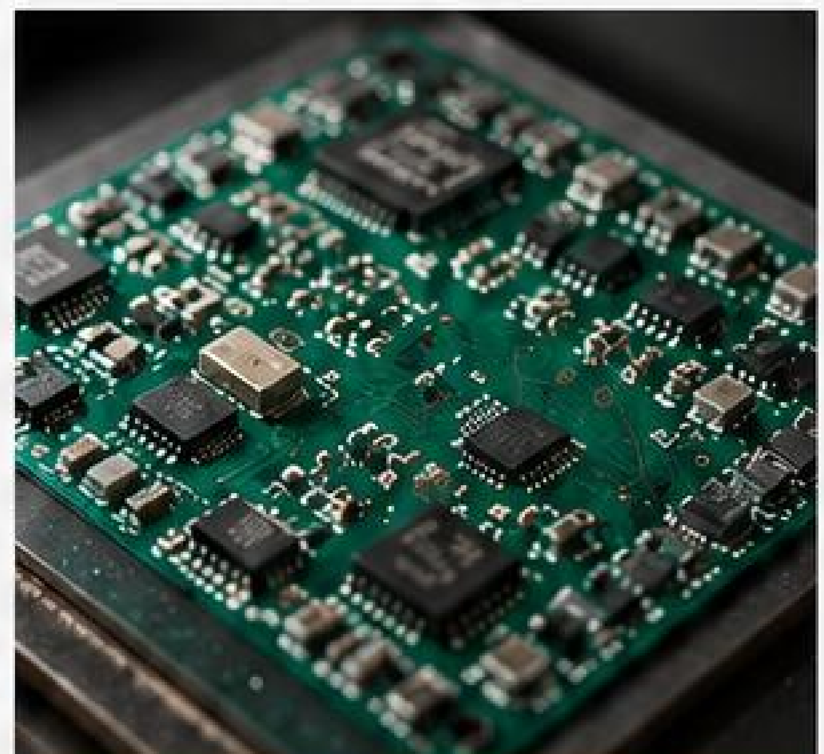
IN-HOUSE ENGINEERING
Complete control from concept to certification of our products.

DESIGNED & MANUFACTURED IN SPAIN
All critical components are developed and produced at our facility in Spain.

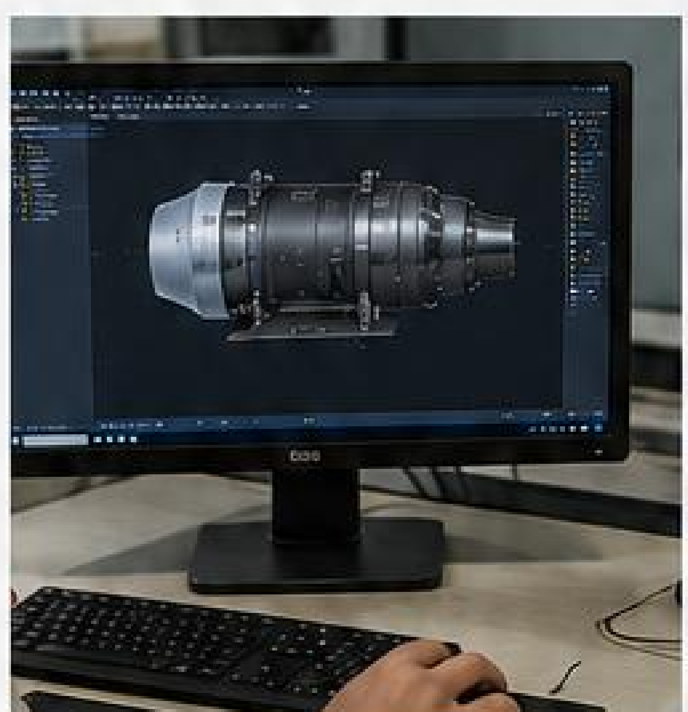
EUROPEAN SUPPLY CHAIN
Reliable, secure and ITAR-free solutions for international customers.

UAV & DEFENSE FOCUS
Engines and systems optimized for UAV, loitering munitions, target drones and special mission platforms.

PRECISION MANUFACTURING



ELECTRONIC CONTROL UNITS



DESIGN & SIMULATION

TESTING & VALIDATION

25+	1000+	20+	AGILE	SHORT
YEARS OF EXPERIENCE	ENGINES DELIVERED	COUNTRIES SERVED	DEVELOPMENT PROCESS	LEAD TIMES

“ Our mission is to deliver compact, reliable and efficient propulsion systems that enable the next generation of unmanned platforms.

RELIABLE PERFORMANCE | FLEXIBLE SOLUTIONS | CUSTOMER FOCUSED

JETSMUNT
DEFENSE PROPULSION SYSTEMS

COMPLETE PRODUCT PORTFOLIO

A COMPREHENSIVE RANGE OF COMPACT TURBOJET ENGINES

Jetsmunt offers a complete family of turbojet propulsion systems, delivering reliable performance for a wide range of UAV platforms and mission requirements.

98 N – 255 N
Thrust range

MULTIPLE UAV CLASSES
From light to heavy platforms

PROVEN TECHNOLOGY
6,000+ engines delivered worldwide

98 N
XM98NG

Diameter: 90 mm
Length: 210 mm
Weight: 968 g
Consumption: 272 g/min

122 N
XM122NG

Diameter: 90 mm
Length: 210 mm
Weight: 978 g
Consumption: 340 g/min

166 N
XM166NG

Diameter: 102 mm
Length: 225 mm
Weight: 1,420 g
Consumption: 408 g/min

182 N
XM182NG

Diameter: 102 mm
Length: 225 mm
Weight: 1,435 g
Consumption: 435 g/min

210 N
XM210NG

Diameter: 110.9 mm
Length: 250 mm
Weight: 1,785 g
Consumption: 560 g/min

222 N
XM222NG

Diameter: 110.9 mm
Length: 250 mm
Weight: 1,800 g
Consumption: 575 g/min

250 N
XM250NG

Diameter: 122 mm
Length: 265 mm
Weight: 2,080 g
Consumption: 700 g/min

255 N
XM255NG

Diameter: 122 mm
Length: 272.6 mm
Weight: 2,080 g
Consumption: 715 g/min

NG SERIES

Reliable and efficient propulsion systems optimized for a wide range of UAV platforms and operational environments.

PRO SERIES

Enhanced integration capabilities including advanced telemetry, electronic control and mission-specific interfaces.

AVAILABLE THRUST CLASSES FROM 98 N TO 255 N

Designed, developed and manufactured in Spain

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XM215 PRO

COMPACT UAV PROPULSION SYSTEM

The XM215 PRO is a compact turbine engine designed for high performance UAVs, loitering munitions and target drones. It delivers reliable thrust in a lightweight, fuel-efficient and robust package, optimized for demanding missions.

FRONT VIEW

3/4 VIEW

REAR VIEW

HIGH PERFORMANCE
215 N thrust with excellent fuel efficiency.

LIGHTWEIGHT DESIGN
Optimized for maximum thrust-to-weight ratio.

RELIABLE & ROBUST
Proven technology for demanding environments.

EASY INTEGRATION
Compact form factor for rapid platform integration.

ENGINE SPECIFICATIONS

Max. Thrust	215 N
Max. RPM	122,000 rpm
Engine Weight	1,820 g
Fuel Pump Weight	65 g
Total Installed Weight	1,885 g
Length	265 mm
Body Diameter	73.9 mm
Max. Diameter	112.5 mm
Height	177.5 mm
Power Supply	12 – 32 V DC

Specifications subject to change without notice.

INTEGRATED ELECTRONICS & CONTROL

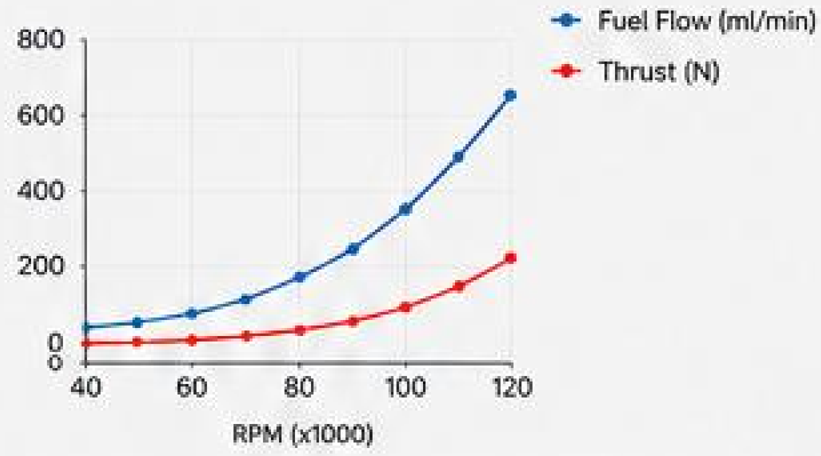
- All electronics integrated inside the engine
- Extended power supply range: 12–32V DC
- Over current and over voltage protection
- Extended operating temperature range: -30 °C to +80 °C
- CAN Bus control and telemetry interface (optional)
- Fiber Optic control and telemetry interface (optional)
- I²C interface for additional sensors (fuel flow, airspeed, telemetry, etc.)

KEY FEATURES

- High reliability electronics
- True sine wave, Field Oriented Control of starter and pump brushless motors
- Ultra precise control down to a few RPM
- Ambient temperature, pressure and humidity sensor
- SD card interface for data logging up to 50 Hz
- Real Optic Kalman filtering of all sensor data for a more precise control loop
- Easy integration with most autopilots via open serial protocol
- Custom data protocol and configuration to customer requirements
- Full access to telemetry data and configuration parameters

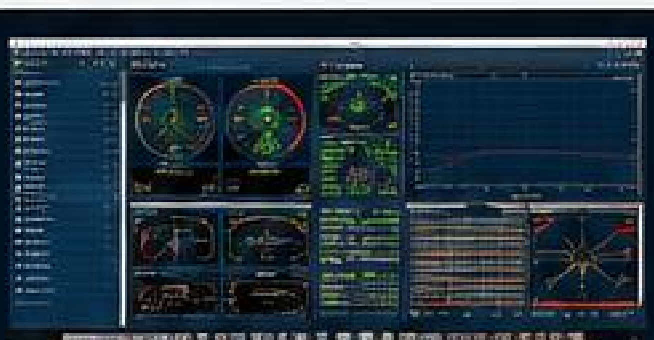
THRUST PERFORMANCE

RPM	Fuel Flow (ml/min)	Thrust (N)
122,000	652	215
112,000	458	161
102,000	366	125
90,000	255	85
80,000	128	68
60,000	92	30
50,000	69	22
40,000	45	14
28,000	33	8



INTEGRATION FEATURES

- Integrated electronics for simplified installation
- Open serial protocol for autopilot integration
- CAN Bus telemetry and control
- Fiber optic interface (optional)
- SD card data logging up to 50 Hz
- Cross-platform configuration and monitoring software



CONFIGURATION & MONITORING SOFTWARE

We provide a powerful graphical interface for configuration, tuning, operation and monitoring of the engine.

Cross-platform and compatible with all major operating systems: Mac OS, Linux and Windows.

macOS Linux Windows

DESIGNED, DEVELOPED AND MANUFACTURED IN SPAIN
From standard engines to custom propulsion solutions.

XM255 PRO

HIGH PERFORMANCE UAV PROPULSION SYSTEM

The XM255 PRO is a high-performance turbojet engine developed for advanced UAV platforms, target drones and special mission applications.

With increased thrust capability, integrated electronics and advanced telemetry architecture, the XM255 PRO provides reliable propulsion for demanding aerospace environments.


FRONT VIEW


3/4 VIEW


REAR VIEW

HIGH PERFORMANCE
255 N thrust with excellent fuel efficiency.

LIGHTWEIGHT DESIGN
Optimized for maximum thrust-to-weight ratio.

RELIABLE & ROBUST
Proven technology for demanding environments.

EASY INTEGRATION
Compact form factor for rapid platform integration.

ENGINE SPECIFICATIONS

Max. Thrust	255 N
Max. RPM	110,000 rpm
Engine Weight	2,080 g
Length	272.6 mm
Max. Diameter	122 mm
Height	166.5 mm
Power Supply	12 – 32 V DC

Specifications subject to change without notice.

INTEGRATED ELECTRONICS & CONTROL

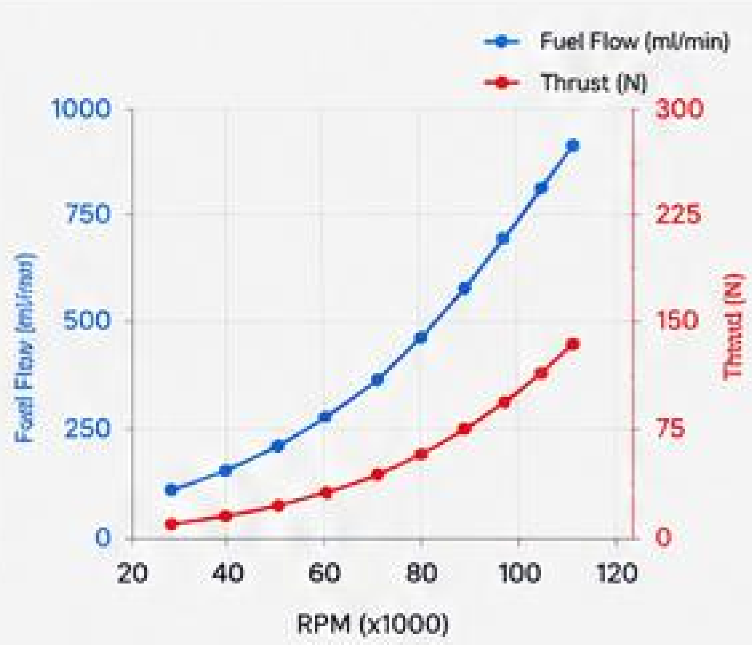
- All electronics integrated inside the engine
- Extended power supply range: 12–32V DC
- Over current and over voltage protection
- Extended operating temperature range: -30 °C to +80 °C
- CAN Bus control and telemetry interface (optional)
- Fiber Optic control and telemetry interface (optional)
- I²C interface for additional sensors (fuel flow, airspeed, telemetry, etc.)

KEY FEATURES

- High reliability electronics
- True sine wave, Field Oriented Control of starter and pump brushless motors
- Ultra precise control down to a few RPM
- Ambient temperature, pressure and humidity sensor
- SD card interface for data logging up to 50 Hz
- Real time Kalman filtering of all sensor data for a more precise control loop
- Easiy integration with most autopilots via open serial protocol
- Full access to telemetry data and configuration parameters

THRUST PERFORMANCE

RPM	Fuel Flow (ml/min)	Thrust (N)
110,000	875	255
100,000	610	196
90,000	439	144
80,000	347	114
70,000	255	72
60,000	202	51
50,000	132	37
40,000	105	26
28,000	52	10



OPERATING CONDITIONS

Range of Starting Conditions

Speed	0 – 450 km/h
Altitude	0 – 6500 m
Temperature	-35°C to +50°C

Range of Running Conditions

Speed	0 – 850 km/h
Altitude	0 – 10,000 m
Temperature	-45°C to +50°C

INTEGRATION FEATURES

- Integrated electronics for simplified installation
- CAN Bus telemetry and control
- SD card data logging up to 50 Hz
- Open serial protocol for autopilot integration
- Fiber optic interface (optional)
- Cross-platform configuration and monitoring software

CONFIGURATION & MONITORING SOFTWARE

We provide a powerful graphical interface for configuration, tuning, operation and monitoring of the engine.

Cross-platform and compatible with all major operating systems: Mac OS, Linux and Windows.

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ADVANCED ELECTRONICS & TELEMETRY

Integrated Propulsion Intelligence

Designed for seamless integration into modern UAV platforms.

INTEGRATED CONTROL ARCHITECTURE

- GROUND CONTROL STATION
- TELEMETRY
- AUTOPILOT
- JETSMUNT ECU
- TURBOJET ENGINE
- SENSORS & ACTUATORS

Bidirectional data and control flow for maximum performance and reliability.

COMMUNICATION INTERFACES

CAN BUS
Reliable high-speed telemetry and engine control.

FIBER OPTIC INTERFACE (OPTIONAL)
Optional interface for advanced UAV architectures.

OPEN SERIAL PROTOCOL
Simple integration with most autopilot platforms.

I²C SENSOR EXPANSION
Support for additional mission sensors.

SD DATA LOGGING
High-frequency recording for analysis and diagnostics.

ADVANCED ENGINE MANAGEMENT

REAL-TIME TELEMETRY
Continuous monitoring of critical engine parameters.

INTELLIGENT SENSOR FUSION
Kalman-filter-based processing for enhanced accuracy.

ENVIRONMENTAL COMPENSATION
Automatic adaptation to pressure, temperature and humidity.

ENGINE PROTECTION FUNCTIONS
Integrated monitoring and protection logic.

PRECISION RPM CONTROL
Stable and accurate turbine operation across the entire envelope.

ENGINE HEALTH MONITORING
Continuous performance supervision and trend analysis.

CONFIGURATION & MONITORING SOFTWARE

We provide a powerful cross-platform software suite for complete engine configuration, monitoring, diagnostics and data analysis.

- Engine configuration
- Real-time monitoring
- Diagnostics & troubleshooting
- Data logging up to 50 Hz
- Performance analysis
- Firmware updates
- Parameter management

CROSS PLATFORM COMPATIBILITY

macOS | Linux | Windows

Our software is designed to run on all major operating systems for maximum flexibility in the field and in the lab.

INTELLIGENT PROPULSION.
MAXIMUM PERFORMANCE.

HIGH RELIABILITY · PRECISION CONTROL · SMART TELEMETRY · SECURE & ROBUST · EASY INTEGRATION

UAV INTEGRATION CAPABILITIES

SEAMLESS INTEGRATION. MAXIMUM PERFORMANCE.

Jetsmunt propulsion systems are engineered for rapid and reliable integration into a wide range of UAV platforms, providing the power, efficiency and reliability required for mission success.

INTERFACES

- **CAN BUS** — SAE J1939 compatible
- **I²C BUS** — For sensor and peripheral communications
- **SERIAL LINK** — RS232 / RS422 / RS485
- **PWM** — For throttle and control signals
- **SD CARD** — For data logging and firmware updates

APPLICATIONS

FIXED WING UAVs
VTOL / HYBRID UAVs
TARGET DRONES
LOITERING MUNITIONS
GROUND-LAUNCHED SYSTEMS
SPECIAL MISSION PLATFORMS

INTEGRATION FEATURES

- **COMPACT & LIGHTWEIGHT** — Optimized for minimal integration impact
- **FLEXIBLE MOUNTING** — Multiple mounting configurations available
- **PLUG & PLAY** — Standardized interfaces for quick integration
- **HIGH RELIABILITY** — Proven in demanding operational environments
- **TECHNICAL SUPPORT** — Dedicated engineering support throughout the program

TYPICAL INTEGRATION ARCHITECTURE

Jetsmunt ECUs provide full authority control, engine monitoring and data logging for seamless UAV platform integration.

KEY BENEFITS

- **REDUCED INTEGRATION TIME** — Standardized interfaces and plug & play design
- **OPTIMIZED PERFORMANCE** — Maximizes thrust-to-weight and fuel efficiency
- **MISSION RELIABILITY** — Built for mission-critical applications
- **COST EFFECTIVE** — Lower development and operational costs
- **GLOBAL SUPPORT** — Worldwide technical and service support

ENGINEERING & CUSTOM DEVELOPMENT

From concept definition to flight-ready propulsion systems.

Jetsmunt supports OEMs, research organizations and aerospace programs requiring propulsion solutions beyond standard catalogue products.

Our engineering team provides development support from concept definition through prototype manufacturing, testing and flight validation.

DEVELOPMENT CAPABILITIES

- Thrust Optimization
- Dimensional Adaptation
- ECU Customization
- Telemetry Integration
- Prototype Manufacturing
- Environmental Qualification
- Flight Test Support
- OEM Engineering Assistance

DEVELOPMENT PROCESS

MISSION REQUIREMENTS → CONCEPT DESIGN → ENGINEERING DEVELOPMENT → PROTOTYPE MANUFACTURING → GROUND TESTING → FLIGHT VALIDATION

A complete development cycle under one roof.
From initial requirements to flight-proven solutions.

AVAILABLE DEVELOPMENT AREAS

COMPACT UAV ENGINES
Adaptation of existing propulsion platforms.

INCREASED THRUST PLATFORMS
Future propulsion developments tailored to mission requirements.

INTEGRATED PROPULSION SYSTEMS
Engine, electronics and telemetry solutions.

SPECIAL MISSION CONFIGURATIONS
Custom integration support for advanced aerospace applications.

WHY JETMUNT

- FOUNDED IN 1998
- MORE THAN 25 YEARS OF TURBINE DEVELOPMENT
- THOUSANDS OF ENGINES DELIVERED WORLDWIDE
- IN-HOUSE ENGINEERING
- IN-HOUSE MANUFACTURING
- EUROPEAN SUPPLY CHAIN
- FAST PROTOTYPING CAPABILITY

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DEFENSE PROPULSION SYSTEMS

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EUROPEAN TURBOJET PROPULSION SYSTEMS

FOR ADVANCED UAV PLATFORMS

Founded in 1998, Jetsmunt designs and manufactures compact turbojet propulsion systems for UAV and defense applications.

With more than 25 years of experience and thousands of engines delivered worldwide, Jetsmunt supports customers with both standard propulsion products and custom development programs.

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FROM STANDARD ENGINES TO CUSTOM PROPULSION SOLUTIONS